

associations with the adoption of computer technology—are as pervasive within gender, education, and occupation groups as between, indicating that these supply side forces are not the primary explanation for our results.

We begin by presenting our informal “task model” describing how computerization affects the tasks that workers and machines perform. We next formalize this model in a production framework to develop empirical implications for task demand at the industry and occupation level. Subsequent sections describe our data sources and test the model’s main implications. Drawing together the empirical strands, we finally assess the extent to which changes in the task composition can account for recent demand shifts favoring more educated workers. This exercise shows that estimated task shifts are economically large, underscoring the potential of the conceptual model to reconcile key facts.

I. THE TASK MODEL

We begin by conceptualizing work from a “machine’s-eye” view as a series of tasks to be performed, such as moving an object, performing a calculation, communicating a piece of information, or resolving a discrepancy. Our model asks: which of these tasks can be performed by a computer? A general answer is found by examining what is arguably the first digital computer, the Jacquard Loom of 1801. Jacquard’s invention was a machine for weaving fabrics with inlaid patterns specified by a program punched onto cards and fed into the loom. Some programs were quite sophisticated; one surviving example uses more than 10,000 cards to weave a black and white silk portrait of Jacquard himself.³ Two centuries later, the electronic descendants of Jacquard’s loom share with it two intrinsic traits. First, they rapidly and accurately perform repetitive tasks that are deterministically specified by stored instructions (programs) that designate unambiguously what actions the machine will perform at each contingency to achieve the desired result. Second, computers are “symbolic processors,” acting upon abstract representations of information such as binary numbers or, in the loom’s case, punched cards.

Spurred by a more than trillionfold decline in the real price of

3. The Jacquard loom was also the inspiration for Charles Babbage’s analytical engine and Herman Hollerith’s punch card reader, used to process the 1910 United States Census.